

# Quasar Identification in J-PAS data using simulations from DESI

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# Quasars, J-PAS Survey & DESI

## What are quasars?

- Supermassive black holes actively accreting matter.
- Extremely luminous, visible across vast cosmic distances.

## The J-PAS survey

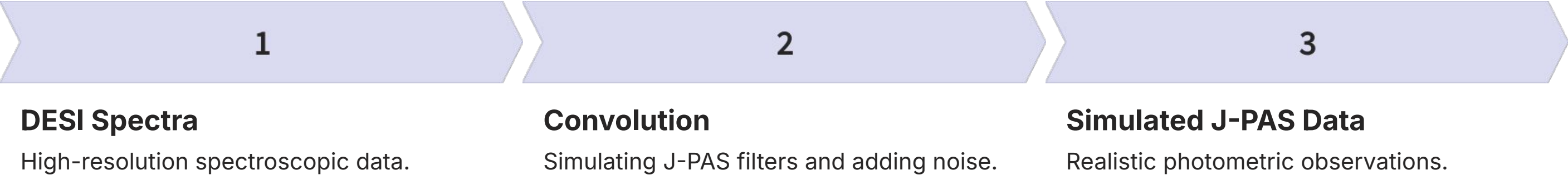
- Photometric telescope using 56 narrow-band filters.
- Produces low-resolution photometric spectra.

## DESI

- Largest existing spectroscopic survey.
- Produces high-resolution spectroscopic data.



# Dataset Generation



**The Dataset:**

**Training:** 3.45M objects  
**Validation:** 740K objects  
**Testing:** 740K objects  
**Real:** 52K objects

**Features:**

- Photometric fluxes (56 bands) + Noise
- RA / DEC (sky position)
- Morphology
- EBV

\***Redshift** included in **features** later to improve model performance.

# Data Pre-Processing

## Feature Selection

Selected relevant fluxes and physical properties as features: 'RA', 'DEC', 'DESI\_FLUG\_G', 'DESI\_FLUX\_R', 'DESI\_FLUX\_Z', 'EBV', 'MORPHTYPE', 'NOISE\_TILE'



## Data Concatenation

Fluxes and metadata combined as model input.

## NaN Removal

Eliminated 0.07% of rows containing NaNs per subset.



## StandardScaler Normalisation

Scaled data to mean 0, standard deviation 1.

# Machine Learning Models Applied



## Random Forest (RF)

Ensemble of decision trees.

Robust to noisy data.



## K-Nearest Neighbors

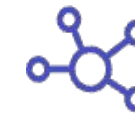
Classifies based on the majority class among the  $k$  closest data points. Sensitive to feature scaling and noisy data.



## Stochastic Gradient Descent (SGD)

Incremental optimisation.

Sensitive to data quality.



## Multilayer Perceptron (MLP)

Neural network with ReLU and Adam.

Captures non-linearities.



# Hyperparameter Optimisation

## GridSearch

Discarded for its slowness.

| Hyperparameter RF | Selected Values         |
|-------------------|-------------------------|
| Max_depth         | None                    |
| Bootstrap         | False                   |
| Min_samples_leaf  | 3                       |
| Min_samples_split | 4                       |
| Cass_weights      | Star 1, Galaxy 1, QSO 4 |

## Optuna

Used for RF and SGD. Fast Bayesian optimisation.

| Hyperparameter SGD | Selected Values         |
|--------------------|-------------------------|
| Loss               | log_loss                |
| Penalty            | elasticnet              |
| Alpha              | 1e-4                    |
| Learning_rate      | Adaptive                |
| Class_weight       | Star 2, Galaxy 1, QSO 6 |

## RandomisedSearch

Used for MLP. Efficient for continuous hyperparameters.

| Hyperparameter MLP | Selected Value |
|--------------------|----------------|
| Hidden Layer Size  | (100,50)       |
| Activation         | Rely           |
| Solver             | Adam           |
| Learning Rate      | Constant       |

# Results Without Redshift

| Model         | Precision | Recall | F1-Score |
|---------------|-----------|--------|----------|
| Random Forest | 0.8067    | 0.7219 | 0.7619   |
| SGD           | 0.4996    | 0.5681 | 0.5316   |
| MLP           | 0.8467    | 0.6701 | 0.7481   |

## Model Performance Summary

- **Random Forest and MLP:** good and robust performance.
- **SGD:** struggled, unstable with ambiguous data.

These results reflect the models' intrinsic capabilities without using additional spectral information.

# Results With Redshift

| Model         | Precision | Recall | F1-Score |
|---------------|-----------|--------|----------|
| Random Forest | 0.9429    | 0.8652 | 0.9024   |
| SGD           | 0.7774    | 0.7656 | 0.7714   |
| MLP           | 0.9528    | 0.8486 | 0.8977   |

## Impact of Adding Redshift

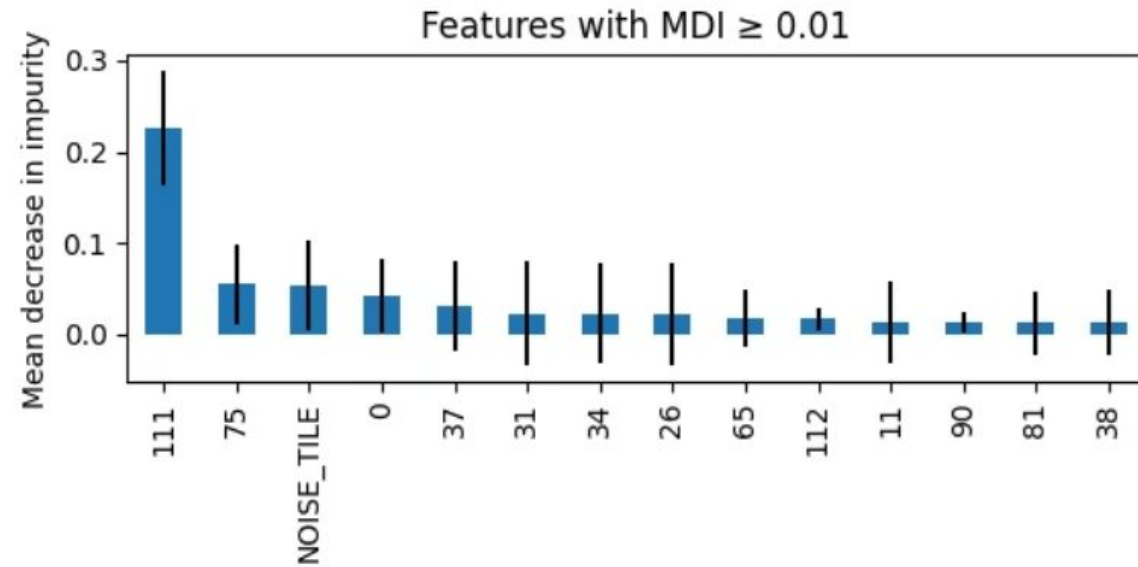
- **F1-scores improved** for all models.
- **Redshift helps separate** galaxies and quasars.
- **Provides indirect spectral information.**

Adding redshift provides extra information that helps better separate quasars and galaxies.

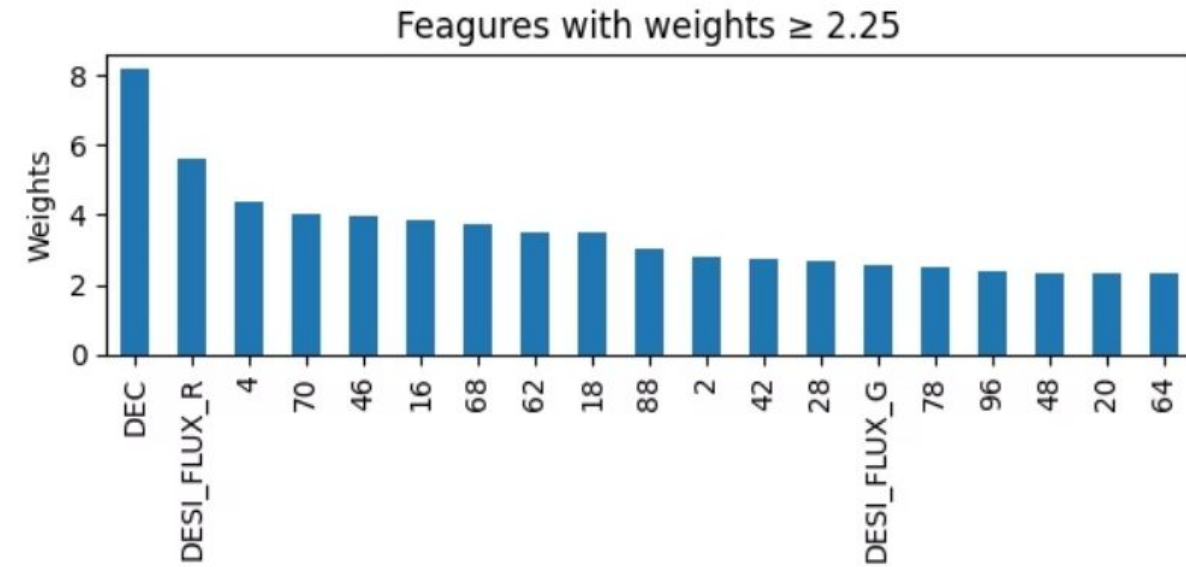


# Feature Importances

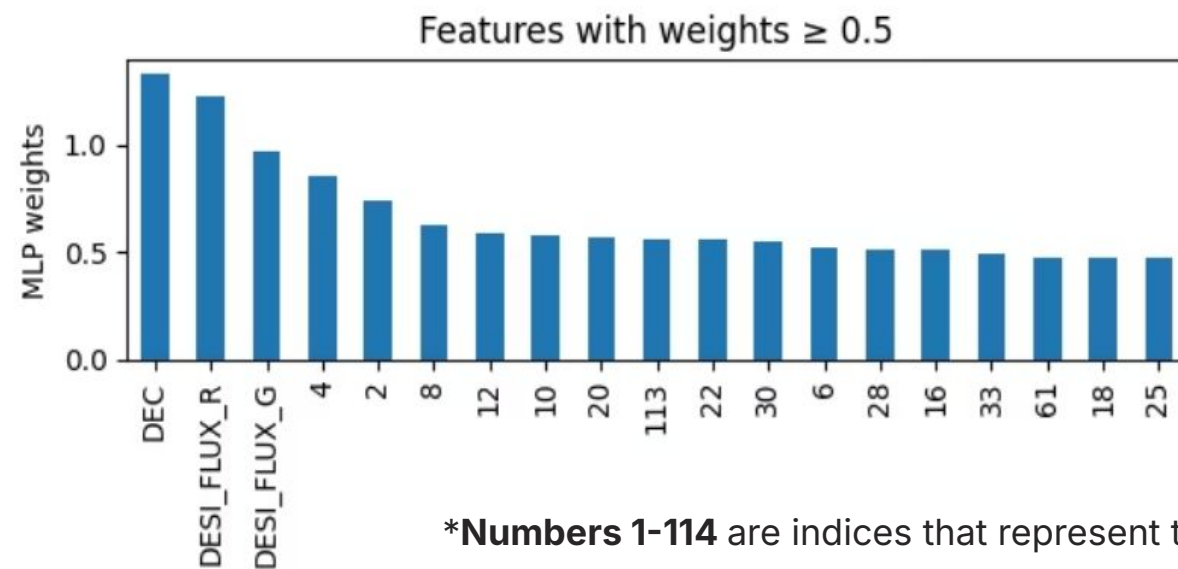
## Random Forest (RF)



## Stochastic Gradient Descent (SGD)



## Multi-Layer Perceptron (MLP)



### Key Points

- **Photometric fluxes** are the most important features.
- Some non-photometric metadata, like **DEC**, is also important.
- MLP and SGD share leading features.

\*Numbers 1-114 are indices that represent the **wavelengths and the fluxes with noise** of each filter

# Conclusions



## Random Forest

Overall best model, robust.



## MLP

Performs well, requires tuning.



## SGD

Struggles with conflicting gradients and ambiguous data.

- Photometric redshift is crucial for performance, but limited by data availability.
- The ability to work with large datasets containing diverse features has been achieved.
- Overfitting and mitigating astrophysical noise is reduced

